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A Cloud Migration Checklist

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Migrating Workloads to Public Cloud

According to a recent JP Morgan survey of more than 200 CIOs of large enterprises: 16.2% of workloads are currently running in the public cloud, and in five years 41.3% of workloads will run in a public cloud. If that's the case in 5 years, what will happen in a decade? As AWS becomes a $10B business, we can definitively say that enterprises are looking to leverage the public cloud.

However, even though self-provisioning of new workloads on AWS or Azure is simple, migration of a running service to the cloud requires a bit more planning. It's a common perception that migrating existing workloads to public cloud—especially those with a lot of data—is complex, time consuming and risky, but technology is rapidly evolving and it's easier than ever to get the job done. In this article, we will provide the essential tips (plus a checklist at the end) to help guide IT organizations through the five key parts of the migration process: Discovery and Assessment, Planning, Pilot and Testing, Migration and Cutover, and Ongoing Operations.

1. Discovery and Assessment: Lay the Migration Groundwork

Before you dive into the details of the migration itself, it's usually helpful to start with something more organic: your team. It is likely you'll be working with other members of your organization, so determine who those stakeholders are now and determine what level of involvement is most appropriate for them throughout. For example, someone on your network security team may want to review your migration plans before you get started. And, your support team may need some additional training once apps have been migrated to the cloud. Layout these stakeholders and their needs beforehand so people are prepared.

Once you've got the people piece figured out, it's time to dive into the migration itself. It's likely that your enterprise has hundreds (or even thousands) of applications to migrate to the cloud, so where do you begin? The first part is understanding which applications can be moved to the cloud. For example, some applications may have Operating System limitations that prevent you from being able to migrate it at all. Prepare a list of your existing application catalog and eliminate anything which simply won't work in the cloud.

But what many companies are realizing is that even very well researched and planned migrations still run into speedbumps, and until the application is live in the cloud, IT managers don't have a clear picture of performance, fit, and cost. For applications with large, dynamic data sets, moving the data first can be time consuming and complicated. Consider tools that will help you test applications in the cloud before you migrate them.

2. Planning: Prepare for Migration

According to Gartner, there are essentially two cloud migration strategies that IT can use:

1. Rehost ("Lift and shift")—redeploy applications on an IaaS without making changes.
2. Revise—modify or extend existing code to fit the cloud environment.

If you are able to test your applications before you migrate them, then you are already aware of dependencies or other adjustments that might need to be made. This makes mass migrations vastly easier, as long as you have migration tools that employ deduplication, WAN optimization and other capabilities to ensure bandwidth is used efficiently. You can then have applications up and running in the cloud, while continuing to implement optimizations over time.

Review the cloud capabilities

Once you've determined your strategy, identify the key areas where you can leverage your cloud provider's services—things like DBaaS (Database as a Service), DNS services or other application-level managed services. Also be sure to review all the cloud environment prerequisites for supporting the migrated workloads like networking (subnets, services etc.), security, and surrounding services.

Plan data migration

Migrating an application often involves migrating data along with it. Consider the amount of data involved and how it is updated, and understand that synchronization with your on-premises data repositories might be required as part of the cutover. Replication-based migration typically includes a number of steps:

1. Replicating some (or all) of the dataset, which can take anywhere between hours to weeks depending on which tools you use.
2. Identifying and synchronizing the delta between the replicated copy (in the cloud) and the production (live) dataset on premises, and moving that before the cutover.
3. Cutover, at which time the workload actually begins running in the cloud.

Alternatively, you may want to consider a streaming-based migration tool, which can move only the necessary elements to get an application up and running (so, cutover first), then migrate the data in the background while applications run, so any new data is already resident in the cloud.

There are two tactics with respect to data migration:

1. Full replication, where you copy the entire data set from on-premises into the cloud in the initial phase of migration. Be aware that this approach can require significant time investment, especially for data sets measured in Terabytes.
2. Streaming, where you copy an initial subset of critical data into the cloud so that your application can begin running. Then, in the background, you continue to upload data into the cloud, as well as keep the on-premises data synchronized with any changes made in the cloud. This typically saves tens or hundreds of hours during a migration project.

Tools and software

Migrating applications doesn't mean you have to manually install and copy/paste things up in to the cloud. There are software tools that will drastically reduce your time, cost, and risk while migrating applications to the cloud. Ideally, you're looking for software which can accomplish the following:

- Easily integrate into your existing on-premises and future cloud provider's software
- Migrate both the application and the application's data seamlessly and quickly into the cloud
- Provide an easy way to roll back the application to on-premises if needed
- Automatically customize and fine tune the migration process and cloud VM instances
- Provide insight into the monitoring process both in-progress and after completion

Document your plan

Take all of the research you've done above and create a blueprint design of the environment in the cloud. Be sure to identify and document all of the requisite building blocks, things like:

- Network topology
- System layers
- Cloud instance configuration
- Migration order
- Dependencies
- Tools / software

3. Pilot: Test Your Workload

During assessment or after, run a proof of concept for each application you are planning to migrate. The pilot project will help you validate two crucial pieces of your migration:

1. The resources and capacity your application requires
2. Your cloud vendor's capabilities and limitations (e.g. number of VMs, storage types and size, and network bandwidth)

A good example where this is crucial is memory-intensive applications. You need to do the proper performance testing by simulating real-world load on the system so you can choose the right sized instance with enough memory to run your application in production. The pilot can also help you understand which storage services will replace your different traditional repositories (e.g., Amazon cloud EFS to replace your traditional NAS appliance).

Test, test and test some more

The more you're able to test in the beginning, the smoother your migration will ultimately go. We advise running tests which validate:

- Workload functionality, performance, and costs when running in the cloud
- Performance which is representative of the final production environment
- Migration tools providing the simplest and most optimal migration

Once you've got your desired setup and migration suite, you should also do one final retest of the migration process itself. This helps ensure you have captured the right configurations (settings, security controls, replacement of legacy firewalls, etc.) and perfected your migration. With that final retest, you can also develop a cost-baseline for what your deployment will cost month over month.

4. Migration: Moving to the Cloud

Use a phased approach for moving workloads to the cloud. After each phase, review the results and adjust your plan for lessons learned if needed. Having software that lets you revert back to the on-premises configuration, if necessary, can be a huge benefit at this phase, too.

If you're leveraging a replication tool for migration, make sure to validate data synchronization and consistency throughout migration. Leveraging a streaming-based migration tool generally helps accomplish this.

5. Don't Forget to Consider Ongoing Operations

Change your operational usage habits to meet the cloud's highly dynamic environment. For example, in the cloud, you have more flexibility to provision and adjust instance type and size based on real demand. Track your cloud vendor's product updates and don't be afraid to experiment as you continue to optimize.

As your migration continues, build the maintenance and transparency layers in parallel so that IT can properly manage things like security, performance, availability (backup and DR) and cost. Learn which management tools are provided by your cloud vendor and learn how to use APIs to tailor your own solutions. Evaluate the marketplace to find 3rd party vendors that can support you with operations such as monitoring, backup automation and cloud firewall management as needed as well.

Summary: Your Cloud Migration Checklist

To help you on your cloud migration journey, we've summarized the ideas above into a cloud migration checklist:

Discovery and assessment:

- ☐ Create a list of your applications—who is using what and how often
- ☐ Which applications are cloud-eligible
- ☐ Which applications are cloud-desirable
- ☐ Understand application interdependencies and network configurations
- ☐ Specify security and compliance requirements
- ☐ Validate SLA and high availability requirements

Planning:

- ☐ Pick a strategy for each application: Rehost or Revise
- ☐ Decide on data migration and synchronization strategy: Replication or Streaming
- ☐ Plan and design the cloud infrastructure including services like networking, security, etc.
- ☐ Determine the amount and size of instances and the right mix of storage
- ☐ List the automation and control tools
- ☐ Plan and document your "Runbook" that includes migration specification and order

Pilot and testing:

- ☐ Test data migration and synchronization
- ☐ Measure performance
- ☐ Validate security controls required
- ☐ Evaluate your cloud footprint costs
- ☐ Document necessary changes to be done as part of the actual migration
- ☐ Plan the time required for application cutover
- ☐ Re-test to make sure you are fully prepared for the cutover including a potential downtime

Migration and cutover:

- ☐ Prepare the team and its leaders
- ☐ Migrate according to the plan and the Runbook created
- ☐ Use a phased approach—implement, review and adjust accordingly
- ☐ If significant diversions from planning are found, be sure to reconsider the next steps before continuing with the migration process

Ongoing operations:

- ☐ Monitor application and cloud usage
- ☐ Implement bursting or scaled-usage to optimize user experience
- ☐ Use a phased approach—implement, review and adjust accordingly
- ☐ Empower IT to successfully manage ongoing operations

Creating an effective cloud migration process for your enterprise requires change and flexibility. Moving enterprise applications to the cloud may involve some challenges—but the right planning combined with the right tools will drastically reduce the risks and simplify the process considerably. And, of course, the long term benefits of migrating applications to the cloud is huge, despite some of those initial challenges!

If you're thinking about migrating your applications to the cloud, one of our cloud migration specialists would love to help provide you with some guidance. You can reach us at info@velostrata.com or by calling 1.408.444.6039

Velostrata Inc.
20 N San Mateo Drive, Suite 4
San Mateo, CA 94401 USA

Phone
+1.408.444.6039

Website
velostrata.com

VELOSTRATA